Credit Card Approval Prediction

Phase 5: Project Development Phase

SmartBridge / SkillWallet Virtual Internship Program

Team Member	Role
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1. Sprint-Wise Delivery Summary

Sprint	Deliverables	Status
Sprint 1	application_record.csv / credit_record.csv loaded; univariate, multivariate, and descriptive EDA compl	Completed
Sprint 2	Duplicate removal, DAYS_BIRTH/DAYS_EMPLOYED sign correction, LabelEncoder on all categorica	Completed
Sprint 3	Logistic Regression, Decision Tree, Random Forest, and XGBoost trained and evaluated on an 80/20	Completed
Sprint 4	Flask app.py built with / and /predict routes; index.html form built and wired to the model; end-to-end t	Completed

2. Key Implementation Details

2.1 Feature Engineering (Sprint 2)

credit_record.csv (many rows per applicant) is grouped by ID to compute OPEN_MONTH, END_MONTHS, WINDOW, EMI_PAID_OFF (months paid off), EMI_PASTDUES (months overdue), and NUMBER_OF_LOANS (total months on file). TARGET is derived as Not Approved (0) if an applicant has 2+ overdue months, Approved (1) otherwise, with a small amount of realistic underwriting noise to avoid the model trivially memorizing the label.

2.2 Model Training (Sprint 3)

Each of the four algorithms is implemented in its own file under models/ (logistic_regression_model.py, decision_tree_model.py, random_forest_model.py, xgboost_model.py), each exposing a single function that accepts pre-split train/test data and returns the trained model plus its confusion matrix, classification report, accuracy, and F1-score. train_model.py orchestrates all four, compares F1-scores, and persists the winner with Pickle.

2.3 Flask Integration (Sprint 4)

app.py loads model.pkl, model_columns.pkl, scaler.pkl, and encoders.pkl once at startup. The / route renders the empty form; /predict collects form values, encodes categorical fields with the saved LabelEncoders, scales numeric fields with the saved StandardScaler, and calls model.predict() / predict_proba() to render the Approved / Rejected result with a confidence percentage back on the same page.

3. Code Organisation

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Credit_Card_Approval_Prediction/
■■■ Dataset/                     application_record.csv, credit_record.csv
■■■ Training/                    Credit_Card_Approval_Model_Training.ipynb
■■■ Flask/
■   ■■■ app.py                    Flask routes & prediction logic
■   ■■■ templates/index.html      Applicant form + result display
■   ■■■ model/                    model.pkl, model_columns.pkl, scaler.pkl, encoders.pkl
■■■ Project_Documentation/       Phase-wise reports (this document set)
```